

ALAR, China

WMO: 517300

Lat: 40.550N

Lon: 81.267E

Elev: 1013

StdP: 89.73

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-15.5	-20.5	0.7	-12.6	-18.7	0.8	-11.6	4.3	-2.3	3.7	-3.5	0.5	220	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	34.7	19.9	33.5	19.3	32.3	18.8	22.6	31.5	21.5	30.5	20.5	29.7	1.7	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	16.3	28.7	18.4	15.0	27.0	17.3	14.0	25.6	72.5	31.6	67.8	30.6	64.2	29.8	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.0	4.8	4.0	-18.7	37.2	2.5	1.3	-20.5	38.1	-21.9	38.8	-23.3	39.5	-25.1	40.4
			-19.0	25.0	2.4	1.8	-20.7	26.3	-22.1	27.3	-23.5	28.3	-25.3	29.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-7.4	-1.1	8.0	15.7	20.6	23.8	24.8	23.7	19.2	11.8	3.3	-4.6		
	DBStd	11.60	3.06	3.76	4.01	3.70	3.21	2.49	2.30	2.40	2.60	3.41	3.39	2.81		
	HDD10.0	1621	539	312	89	4	0	0	0	0	0	21	203	454		
	HDD18.3	3161	797	545	321	93	16	1	0	0	21	203	452	712		
	CDD10.0	2179	0	0	26	176	329	413	458	425	276	77	1	0		
	CDD18.3	677	0	0	0	14	87	163	199	167	46	1	0	0		
	CDH23.3	8041	0	0	6	306	1101	1885	2279	1829	589	47	0	0		
	CDH26.7	3343	0	0	0	79	407	836	1074	785	158	4	0	0		
(14) Wind	WSAvg	1.5	1.0	1.2	1.6	2.0	2.1	1.9	1.7	1.6	1.4	1.1	0.9	1.0		
(15) Precipitation	PrecAvg	61	2	1	1	2	7	12	12	12	7	2	1	1		
	PrecMax	106	10	10	11	8	32	34	29	55	32	22	7	14		
	PrecMin	26	0	0	0	0	0	1	1	2	0	0	0	0		
	PrecStd	21	2	2	2	2	8	8	7	12	9	4	2	3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	12.1	23.9	31.4	34.1	35.5	36.8	35.6	31.8	26.9	18.4	6.9		
		MCWB	-0.9	3.9	9.9	14.1	15.7	18.9	20.7	21.4	18.5	13.8	7.7	1.5		
	2%	DB	2.4	10.0	21.2	29.1	32.4	34.1	35.1	33.8	30.1	24.5	15.7	4.4		
		MCWB	-2.3	2.8	8.7	13.0	15.2	18.4	20.9	20.7	17.5	12.7	6.4	-0.4		
	5%	DB	0.6	8.3	19.0	27.1	30.9	32.9	33.6	32.4	28.5	22.4	13.5	2.5		
		MCWB	-3.4	1.8	7.9	12.1	14.9	18.3	20.4	20.3	17.1	11.7	5.6	-1.5		
	10%	DB	-1.1	6.5	16.7	24.8	29.2	31.5	32.1	31.0	26.9	20.3	10.6	0.7		
		MCWB	-4.4	0.6	6.9	11.1	14.3	17.8	20.0	20.0	16.6	10.8	4.3	-2.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	4.7	10.4	14.7	18.0	22.3	24.4	24.2	21.6	15.3	8.3	2.1		
		MCDB	4.4	10.5	22.5	30.1	29.2	32.0	33.4	32.0	28.8	22.2	17.2	6.4		
	2%	WB	-2.2	3.1	9.1	13.4	16.4	20.4	22.6	22.7	19.6	14.0	6.7	-0.1		
		MCDB	2.2	9.4	20.0	27.9	29.5	31.1	32.0	31.2	27.2	22.3	14.9	3.9		
	5%	WB	-3.2	2.1	8.1	12.5	15.5	19.3	21.5	21.5	18.3	12.8	5.6	-1.4		
		MCDB	0.4	7.9	18.2	25.9	28.5	30.0	31.0	30.1	25.9	20.9	13.0	2.1		
	10%	WB	-4.3	0.8	7.1	11.5	14.8	18.4	20.6	20.5	17.2	11.5	4.5	-2.5		
		MCDB	-1.2	6.1	16.2	23.7	27.2	29.0	30.0	29.2	24.9	18.7	10.3	0.5		
(35) Mean Daily Temperature Range	MDBR		13.3	13.8	13.7	14.2	13.9	13.4	13.0	13.2	14.9	17.2	14.7	11.7		
	5% DB	MCDBR	15.3	15.6	16.8	18.8	17.4	16.2	16.2	16.1	17.7	19.3	18.2	13.7		
		MCWBR	11.0	9.1	7.2	6.6	5.3	5.6	6.1	7.0	7.9	8.5	9.6	9.3		
	5% WB	MCDBR	14.8	14.9	16.1	17.1	15.5	14.4	14.2	14.7	15.6	17.6	17.9	13.2		
		MCWBR	10.7	8.8	7.0	6.2	5.3	6.0	6.7	7.7	8.0	8.1	9.5	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.361	0.426	0.538	0.601	0.628	0.660	0.610	0.657	0.660	0.547	0.447	0.385		
	taud		2.033	1.894	1.654	1.558	1.561	1.546	1.661	1.569	1.527	1.672	1.855	1.964		
	Ebn at Noon		752	750	701	694	689	668	698	645	596	613	634	676		
	Edh at Noon		118	159	226	266	271	276	243	260	255	197	142	118		
(44) All-Sky Solar Radiation	RadAvg		2.45	3.38	4.63	5.77	6.56	6.81	6.62	5.91	5.06	4.05	2.84	2.20		
	RadStd		0.29	0.25	0.26	0.25	0.28	0.27	0.27	0.35	0.19	0.16	0.14	0.20		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.42	-1.11	-1.20	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	-0.76	-0.81	N/A	N/A	-51

Nomenclature: See separate page